KnowUrMedicine website

Mumbai teen helps those who can't read

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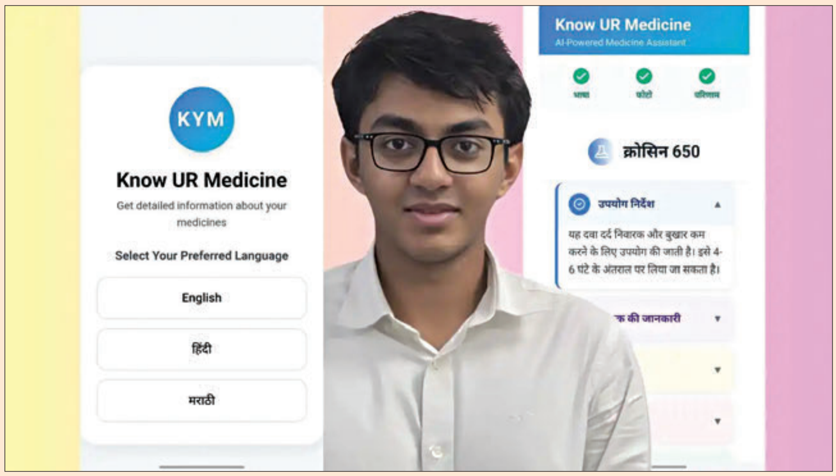
In a Mumbai household, 17-year-old Som Sengupta watched his family's house help bring him an entire medicine box, saying, "Aap hi lelo, Som," unable to parse the English labels to pick the right one. This moment, two years ago, sparked KnowUrMedicine, a web app designed to make medicine information accessible to India's vernacular-speaking and less literate populations.

With 85% of Indians lacking English proficiency and 90% of medicine labels in English: a study shows 45% of patients misinterpret dosage or frequency without native-language instructions, risking health hazards. Som's tech-driven solution leverages AI, optical character recognition (OCR), and text-to-speech (TTS) to deliver critical medicine details in languages like Hindi and Marathi, all through a scalable web platform. This 1000-word feature dives into the technical ingenuity behind Som's altruistic innovation.



A scalable web solution

"I wanted to keep it simple, scan a QR code, select a language, and get the info," Som explains. The backend uses a lightweight framework to handle processing efficiently. Som avoided complex frameworks, opting for a simpler ChatGPT 4o based solution to streamline development. It works entirely in the cloud, avoiding local storage to maintain user privacy and scalability. "No user data is stored," Som emphasizes.

The core functionality hinges on three technologies: OCR to extract text from medicine labels, AI for translation, and TTS for audio output. Users access the app via a QR code or link, choose from Hindi, or Marathi (with plans for more languages), and upload a photo of the medicine. The app processes the image, extracts details, and plays them aloud automatically.

Mastering OCR

The Optical Character Recognition (OCR) system, driven by ChatGPT's API, is the app's backbone and Som's proudest achievement. "Getting the OCR right felt like a breakthrough," he says. He fine-tuned ChatGPT with precise prompts to ensure it recognizes only medicine packaging, rejecting irrelevant inputs like a talcum powder box.

Without using retrieval-augmented generation (RAG) or custom image datasets, Som relied on prompt engineering to filter out "garbage" inputs. "I didn't feed it RAG images; it's just the ChatGPT API with carefully crafted prompts," he says.

Bridging the language divide

Translation is handled by an AI model, which converts English medicine details into Hindi and Marathi. Som avoided building a proprietary medical lexicon, opting for an off-the-shelf AI translator to keep development lean. "The AI handles the heavy lifting," he says, ensuring accurate translations of technical terms like dosage instructions. Early mistranslations were a challenge, but refining the AI prompts improved precision.

Development wasn't without roadblocks. Beyond OCR and TTS issues, Som faced the learning curve of mastering FastAPI and AI integration. "The frameworks took time to understand," he says, crediting his mentor, Sandeep Mistry, for guidance on OCR and pharma outreach. Chemists provided real-world feedback, ensuring the app met practical needs. "They said it helped customers and saved time," he notes, a validation of his technical choices. d>